www.tntextbooks.in

- Primary reproductive organs include the gonads (Testes in male and Ovaries in female).
- Accessory sex organs
  - **Male:** Vas deferens, epididymis, seminal vesicle, prostate gland and penis.
  - **Female:** Fallopian tubes, uterus, cervix and vagina.

The secondary (accessory) sex organs include those structures which are involved in the

- Process of ovulation
- Fusion of the male and female gametes (fertilization)
- Division of the fertilized egg upto the formation of embryo
- Pregnancy
- Development of foetus
- Child birth.

Now let's see the cells of the primary reproductive organs in human male and female and their role in reproduction.

17.7.1 Male Reproductive Organ - Structure of Testes

Testes are the reproductive glands of the male that are oval shaped organs which lie outside the abdominal cavity of a man in a sac like structure called **scrotum**. Now we shall study the various cells which are present in the testes.

Each testes is covered with a layer of fibrous tissue called **tunica albuginea**. Many septa from this layer divide the testes into pyramidal lobules, in which lie seminiferous tubules, cells of Sertoli, and the Leydig cells (interstitial cells).

The process of **spermatogenesis** takes place in the **seminiferous tubules**. The **Sertoli cells** are the supporting cells and provide **nutrients** to the developing sperms. The **Leydig cells** are polyhedral in shape and lie between the seminiferous tubules and secrete **testosterone**. It initiates the process of **spermatogenesis.**

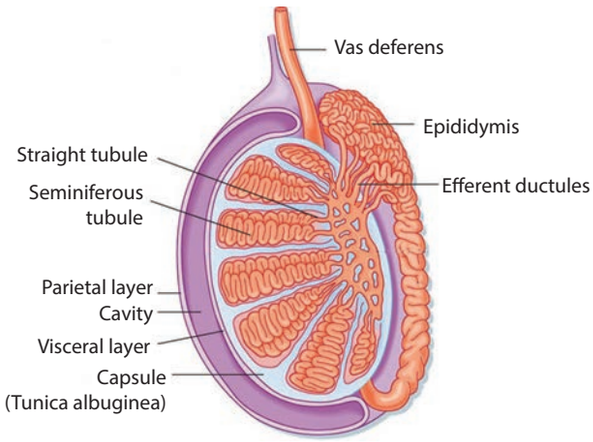


Figure 17.13 L.S of human testes

17.7.2. Female Reproductive Organ - Structure of Ovary

The ovaries are located on either side of the lower abdomen composed of two almond shaped bodies, each lying near the lateral end of fallopian tube. Each ovary is a compact structure consisting of an outer cortex and an inner medulla. The cortex is composed of a network of connective tissue called as stroma and is lined by the **germinal epithelium**. The epithelial cells called the **granulosa cells** surround each ovum in the ovary together forming the primary follicle. As the egg grows larger, the follicle also enlarges and gets filled with the fluid and is called the **Graafian follicle**.

Info bits

The number of primordial follicles in new born female child ranges over 7 million and during reproductive period (at puberty) the number is around 60,000 to 70,000. During a woman's lifetime, she will only ovulate 300 to 400 of the 1-2 million eggs, she was initially born with. On the other side, men will produce over 500 billion sperms in their lifetime.

10th_Science Unit-17.indd 249 06-12-2021 18:43:13